



TEXAS A&M
U N I V E R S I T Y[®]

CSCE 679: Final Report

Ayaan Omair, Phuc Nguyen, Lauren Fuller, Hayley Hawkins

April 24, 2025

1 Introduction:

Heart disease, kidney disease, and cancer are among the most fatal and widespread ailments, devastating the lives of millions of people worldwide every year. While medical treatments have advanced tremendously over the past few decades, it is still important to understand the underlying causes of these conditions. Scientists have been seeking to better understand the complex interplay between biological/genetic factors (gender, race, age, etc.) and lifestyle choices (physical activity level, smoking, consuming alcohol, etc.) in determining the risk of these diseases. For our final project, we have decided to analyze a dataset in the healthcare field. This dataset specifically looks at the key indicators of heart disease. The dataset [1] specifically looks at the key indicators of heart disease. The dataset, titled “Indicators of Heart Disease”, comes from the Center for Disease Control and has information on the main risk factors related to heart disease. The data is a major part of the Behavioral Risk Factor Surveillance System (BRFSS), which conducts annual telephone surveys to collect data on the health status of U.S. residents. This dataset was condensed from 300 variables to 40 variables, and includes key indicators (BMI, gender, age, race, sleep, etc.) relating to health disease. Collecting and analyzing data related to heart disease is crucial for a number of different reasons. It is important in identifying preventative measures to reduce the risk of heart disease for patients in the future. If the dataset is used for research, predictive modeling, or clinical decision-making, the primary audience would be doctors, public health officials, epidemiologists, and medical researchers.

2 Research Questions

This project aims to examine the relationship between lifestyle choices, personal health behaviors, and demographic characteristics and their association with the risk and occurrence of chronic diseases, including heart disease, kidney disease, and skin cancer. The research questions outlined below serve as a framework for investigating patterns and correlations within the data, aiming to uncover broader insights into public health and disease prevalence.

1. Which factors have the strongest correlation with Heart Disease, Kidney Disease, and Skin Cancer?
2. How do lifestyle factors (smoking, alcohol, physical activity) influence disease occurrence?
3. Are certain demographics (men, women, race, age) more prone to getting certain diseases?
4. What are the most common disease co-occurrences?
5. How does sleep duration differ among people with these diseases?

3 Related Work

Heart disease and cancer are among the most fatal and widespread ailments, devastating the lives of millions people worldwide every year. While medical treatments have advanced tremendously over the past few decades, it is still important to understand the underlying causes of these conditions. Scientists have been seeking to better understand the complex interplay between biological/genetic factors (gender, race, age, etc.) and lifestyle choices (physical activity level, smoking, consuming alcohol, etc.) in determining the risk of these diseases. In a recent study, researchers found that certain types of genes are more prone to contracting heart diseases and cancer [2]. As a result, more occurrences were documented in certain ethnicities and genders. In addition to biological elements, lifestyle factors also have a great effect on preventing heart disease and cancer. This is optimistic because while genetic factors are unchangeable at birth, healthy lifestyle habits can be developed and improved. For example, researchers found that a heart healthy diet and regular physical activity significantly reduce cardiovascular risk in the long term [3]. In conclusion, both genetic and lifestyle factors contribute to the likelihood of heart diseases and cancer. While genetic predispositions tend to increase the risk, healthy lifestyle habits help reverse and diminish that risk. Ultimately, modern scientific research will open up new ways to combat these conditions through personalized prevention strategies. With the advent of machine learning, heart disease and cancer treatment has become increasingly effective. Reliable and accurate risk prediction helps monitor a patient's health, often slowing down disease progression, or in some cases even prevents the disease from happening altogether. Traditional algorithms such as logistic regression or decision trees have already achieved remarkable results, setting standard benchmarks in healthcare procedure [4]. Furthermore, as more data becomes available, machine learning models continuously evolve, making them even more robust than ever. As a result, these technologies not only enhance diagnosis accuracy but also assist in medical treatment.

4 Contributions

Our project offers a novel contribution by examining the combined effects of biological and lifestyle factors on disease risk. While much of the existing literature tends to isolate these factors, we take a holistic approach that considers their interactions. This allows us to better understand how lifestyle choices may amplify or reduce risks associated with genetic predispositions, which is a focus of previous studies. Our project uses a reliable, well-curated dataset to offer a strong foundation for meaningful and novel analyses. Our project also uses proven data science techniques for statistical analysis and visualization, solidifying the reliability of our results. In addition, our project is unique because it leverages web development tools to provide an interactive space for our intended audience: researchers, healthcare professionals, or even patients.

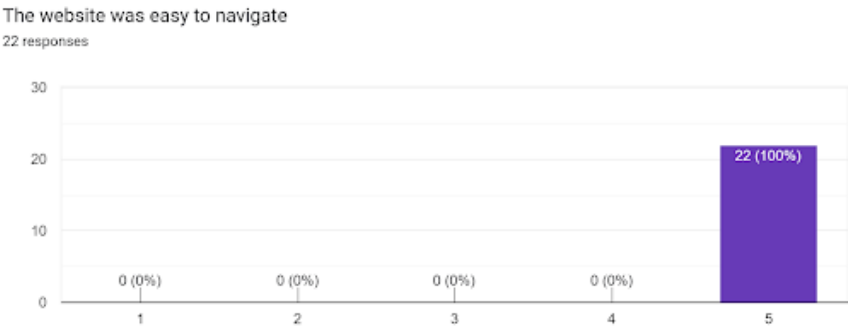
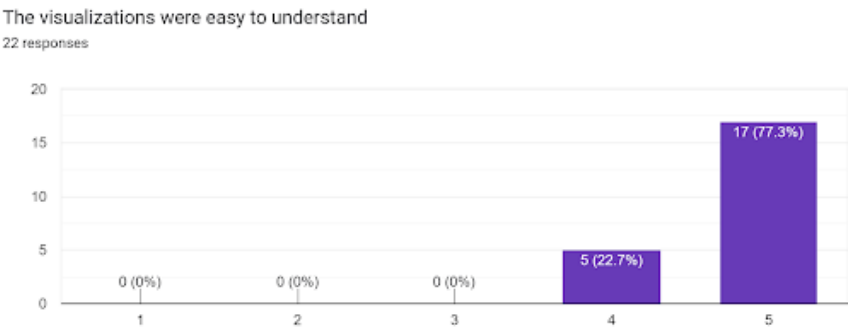
5 Requirement Analysis - Survey Results

Before developing the website, we gathered input from potential users to understand their expectations for visualizing heart disease data. Participants emphasized the importance of clear, interactive visualizations that could highlight correlations between lifestyle factors

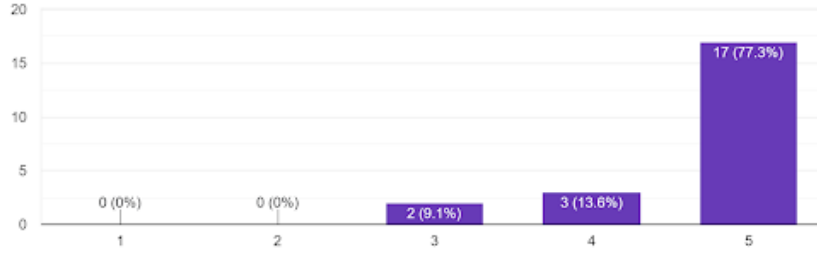
and heart disease, kidney disease, or skin cancer risk. They also expressed interest in tools that allow for personal exploration, such as customizable charts and risk predictors. This feedback directly influenced our design choices, guiding us toward a user-focused, data-driven interface.

Our website feedback centers on a short Google form survey located on our website. The survey has five questions that ask users to rate their experience on the site and describe the aspects that piqued their interest most. Including this survey is a crucial component of our project, as it allows direct input from our audience and assesses the effectiveness of the site’s design, structure, and content delivery. In a research context, this feedback helps ensure that the information is communicated clearly and remains engaging to diverse users. Additionally, the data collected can guide future improvements, helping better tailor the presentation to support educational goals, accessibility, and user engagement. By inviting users to reflect on what they found most compelling, we also gain insight into which parts of the research resonate most with the public.

While survey responses are still being collected, we have already received valuable feedback on our website during its development phase. Initial responses indicate a generally positive user experience. On average, users rated their overall satisfaction at 4.59 out of 5, reflecting strong engagement and appreciation for the content. The clarity of the visualizations received an impressive average score of 4.77 out of 5, suggesting that the data is presented in an effective and easily understandable manner. Navigation received a perfect score of 5 out of 5, indicating that users encountered no difficulties in exploring the website.

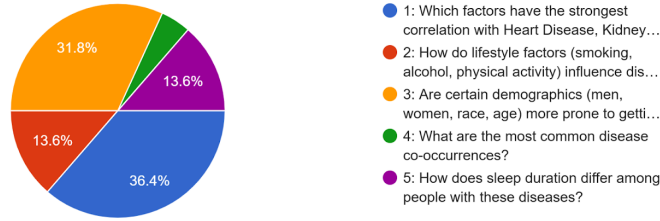


How satisfied are you with your experience using the website
22 responses



Several users specifically mentioned interest in Research Question 1, which explores the factors most strongly correlated with Heart Disease, Kidney Disease, and Skin Cancer, indicating this area may be particularly compelling to visitors. However, multiple respondents also pointed out that some pages are currently empty and that one of the tree visualizations is not fully visible on the site. These comments highlight opportunities for improvement in content completeness and visual accessibility during the site’s development phase.

Which Research Question did you find the most interesting?
22 responses



In summary, while the overall feedback is highly positive, with users appreciating the research direction and visual clarity, there are clear areas to refine as we continue developing the website.

6 System Workflow Overview

The workflow of our system moves from data analysis to web publishing in a streamlined, modular pipeline. The process begins with data preprocessing and statistical analysis in Python, where we explore trends and calculate correlations between variables using techniques such as regression modeling and point-biserial correlation. Once we derive meaningful insights from the data, we export these results into structured formats like CSV or JSON to be used in the visual layer. Next, we transition to building the visualizations in JavaScript using Observable notebooks, where charts are created with D3.js and Observable Plot. These visualizations are built to be interactive, allowing users to filter data by demographic attributes such as gender, race, and age. When filters are applied, the charts update dynamically to reflect the new data selections. Where applicable, the

Python-generated statistical results from the CSV or JSON files (e.g., regression predictive models, t-statistics, or p-values) are visually integrated into the charts, adding analytical depth to the interactive experience. The visualizations are then embedded directly into our website using `<iframe>` elements in the `viz.html` file. This allows seamless integration of live, interactive charts into the website, without needing to manually recreate them in JavaScript within the HTML files. Each HTML file (e.g., `home.html`, `research.html`, `findings.html`, etc.) plays a specific role in presenting the broader narrative of the project. These pages are styled with their own CSS file to ensure a clean, responsive design. Once the site is complete, all files (HTML, CSS, JavaScript, and Python-generated data files) are uploaded to a GitHub repository and hosted through GitHub Pages (<https://lauren-fuller.github.io/679-Final-Project-Webpage/>), making the project publicly accessible without the need for a back-end server infrastructure. This workflow allows for a smooth transition from analysis to interactive presentation and public sharing.

Workflow Diagram:

Data Prep → Python Analysis/Data Cleaning → Python export (JSON or CSV) → Observable Notebook Data Visualization (Javascript) → Embed Visualization into Website via `<iframe>`

7 Algorithm Design

The algorithmic design of this project consists of two primary components: the statistical analysis performed in Python and the reactive, user-driven logic implemented in the visualizations using JavaScript and Observable notebooks.

Statistical Analysis Algorithms

To compute and examine correlations between various factors and diseases, we use a point-biserial correlation coefficient, a statistical measure to determine the strength and direction of a relationship between a continuous variable and a categorical variable. For each disease-outcome pair, we calculate the standardized difference in means between the "disease present" and "disease absent" groups, normalize this by the pooled standard deviation, and apply a D3 linear scale function to map the result into the range $[-1,1]$. This rescaling enhances interpretability and enables consistent visual comparison. We also conducted statistical tests (such as t-tests) to examine mean differences between groups, particularly in metrics like average sleep time. These results inform our visualizations, highlighting if we observed statistically significant differences ($p < 0.05$). The disease prediction tool allows users to input personal information such as BMI, physical activity, general health, and demographic details. These inputs are normalized and converted into a feature vector, which is then passed through a logistic regression function. Internally, each factor is multiplied by a predefined weight, derived from observed correlations in the data, and the resulting weighted sum is passed through a sigmoid function to produce a probability value between 0 and 1. If the predicted probability for a given disease exceeds a threshold, the output indicates a high likelihood "Yes". Otherwise, it predicts "No." This model offers a real-time, interpretable risk estimate based on user input, and is intended as an educational tool to explore how individual traits and habits may relate to chronic disease risk.

Visualization Logic Algorithms

In addition to the backend statistical analysis, we implement an algorithmic structure in our front-end visualizations using JavaScript within Observable notebooks with D3.js. These notebooks listen for user input events (such as selecting a gender or age group), apply filters to the dataset, and re-render the charts accordingly. This reactive algorithm flow is powered by functions like *getSelectedValues()* and *updateChart()*, which dynamically adjust the visual outputs based on user selections. The end result is an easy-to-use experience where users can explore the data themselves and interact with the visualizations to make sense of the patterns we found through the Python analysis. Below, we detail the specific algorithmic logic behind each of the five core visualizations:

Charts 1 and 4 (RQ1 and 4): Correlation Matrices of Diseases and Risk Factors

To evaluate relationships between risk factors and chronic diseases, we implemented a point-biserial correlation algorithm using the custom Observable function:

`pointBiserialCorrelation(data, target, variable)`. This function calculates the correlation between a binary disease variable (e.g., `heartDisease`) and a continuous or encoded categorical variable (e.g., `BMI`, `sleepTime`, or `ageCategory`). Categorical variables such as `race` and `general health` are encoded using the `encodeCategory()` function, which maps category labels to integer indices using `Array.indexOf()`. Within the correlation function, `d3.mean()` and `d3.deviation()` are used to compute group-specific statistics, and the resulting correlation value is scaled to the $[-1, 1]$ range using `d3.scaleLinear()` through the helper function `scaleCorrelation()`. The final matrix is constructed by iterating through every factor and disease target, storing the results in an array of `{target, factor, correlation}` objects, which are later rendered as a heatmap. .

Chart 2 (RQ2): Bar Charts of Lifestyle Factors and Disease

To assess the impact of behaviors like smoking, alcohol consumption, and physical activity, we built a modular data processing function named `chartData(disease, behavior)`. This function uses `d3.rollup()` to group the dataset by disease status and behavioral status (e.g., "HeartDisease" and "Smoking"), then computes the percentage of people in each subgroup using `d3.sum()` and simple division. The output array consists of records with fields such as `percent` and `count`, which represent the relative proportion of each combination. These results are passed into the chart-rendering function `groupedBarChart()`, which generates side-by-side bars that allow visual comparison across all behavior-disease pairs. The chart dynamically updates through Observable's reactive `chartSelector` and `chartDisplay` cells, allowing the user to toggle between nine predefined chart configurations via a dropdown.

Chart 3 (RQ3): Bar Chart of Demographic Filter and Prevalence

To explore how gender, race, and age influence disease rates, we created custom multi-select checkbox widgets using the function `exclusiveCheckboxInput()`. This function applies event listeners to enforce exclusivity between "All" and the specific demographic options. Selected values populate Observable cells such as `selectedGenders`, `selectedRaces`, and `selectedAges`, which are used in `filteredData` to filter the dataset with `.filter()`. The prevalence of six disease types is calculated in `diseaseCounts`, which loops over each binary

outcome and computes the proportion of "Yes" responses within the filtered dataset. When a grouping attribute (e.g., "Race") is selected via `stackBy`, the chart cell dynamically aggregates prevalence rates per subgroup using `Array.flatMap()` and computes percentages for stacked visualization. These values are rendered using `Plot.barY()` in either grouped or stacked format.

Chart 4 Pt. 2 (RQ4 Pt. 2): Disease Co-Occurrence Tree

To visualize common patterns of co-occurring diseases, we constructed a hierarchical tree based on user diagnoses. This logic is implemented in `diseaseTreeData`, where we first filter individuals who report having two or more of the tracked diseases (e.g., HeartDisease, Stroke, Asthma). For each person, the matching disease labels are sorted into a canonical order and used to walk a nested tree structure rooted at `name: "All", count: 0, children: []`. At each level, we check whether a child node already exists using `.find()`, and either increment an existing node's count or create a new one. This results in a trie-like structure that represents shared disease combinations across the population. The resulting tree is rendered using `d3.hierarchy()` and `d3.tree()` in the `diseaseTreeChart` cell, with links generated using `d3.linkHorizontal()`.

Chart 5 (RQ5): Sleep Time Comparison

Sleep duration differences between diseased and non-diseased populations are analyzed using precomputed summary statistics in a python file (`chart5_stat_analysis.ipynb`). The `sleepData` cell contains average sleep hours for people with and without each disease, while the `statsData` cell includes t-statistics and p-values derived from two-sample t-tests performed in Python. These metrics are visualized in a small-multiple bar chart within `chartGrid`, where each disease has its own subplot showing sleep duration bars for both groups. A dynamic narrative is produced in the summary cell using conditional logic: if the p-value is below 0.05, the function concludes that the difference is statistically significant and reports whether those with the disease sleep more or less on average. This conditional explanation is updated live based on user selection in the `diseaseSelect` dropdown.

8 Interface Design

The website's interface is designed to be both functional and user-friendly, guiding visitors through research questions, data findings, and interactive visualizations. The website is composed of six main HTML pages, `home.html`, `research.html`, `viz.html`, `findings.html`, `predictor.html`, and `survey.html`, each fulfilling a distinct purpose in the user journey. A fixed navigation bar at the top of each page allows users to easily switch between sections, creating a smooth and intuitive browsing experience. The interface begins with `index.html`, a simple redirect page that immediately routes users to the main landing page, `home.html`. This homepage includes a navigation menu and introductory content, styled for a clean and welcoming interface. Navigation is uniform across all pages, offering links to Home, Research Questions, Visualizations, Findings, Predictor, and Survey sections. The `research.html` page presents five research questions, each enclosed in individually styled sections. These questions guide the analytical focus of the website. The `viz.html` page complements this by displaying interactive data visualizations sourced from Observable notebooks using javascript. Each

section on this page has a research question with a corresponding visualization embedded from the Observable notebook via `<iframe>`. This allows dynamic exploration of the data through filters and interactive controls provided by Observable’s JavaScript-based platform. The `findings.html` file presents conclusions and analytical summaries derived from the data visualizations featured on the website. The `predictor.html` file contains an interactive questionnaire and visual report powered by a logistical regression model that predicts the user’s risk for developing a disease. The `survey.html` page features an embedded Google Form designed to collect user feedback or supplementary data, with an additional external link for accessibility. Styling across the site is handled through modular, page-specific CSS files (e.g., `home.css`, `research.css`, `viz.css`, etc.). While each page has its own stylesheet for flexibility, global consistency is maintained through shared classes such as `.viz-header`, `.row`, and `.section`. Layouts are built using a combination of custom Flexbox structures and Bootstrap 5’s grid system, enabling responsive behavior across screen sizes.

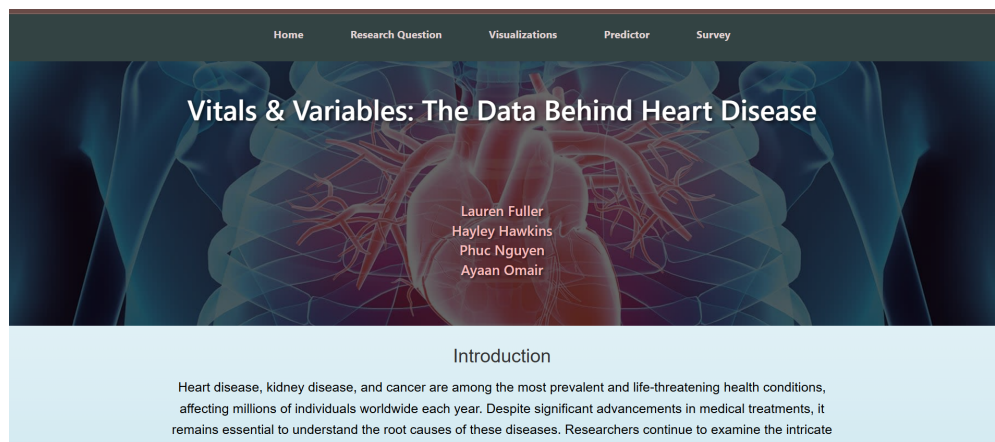


Figure 1: Screenshot of Homepage

9 Code Description

The project brings together multiple functions to create a responsive and interactive data visualization website. At the structural level, the website is built using HTML5, which defines the layout of each page through semantic tags including `<nav>`, `<main>`, `<section>`, `<div>`, and `<iframe>`. These elements help organize the content into logical sections and provide a consistent framework across all five primary HTML files. CSS3 is applied through custom stylesheets that use layout techniques such as Flexbox to align content and maintain responsive behavior across different screen sizes.

In addition, the use of Bootstrap 5 allows for spacing, typography, and flexible layouts in the webpages. In addition to the general architecture, each visualization module is implemented using specialized Observable code tailored to the research question it addresses. Below, we detail the specific code functions used to implement the chart in each of the five core visualizations:

Charts 1 and 4 (RQ1 and 4): Correlation Matrices of Diseases and Risk Factors

These correlation matrices are rendered as heatmaps using `d3.create("svg")`, where each cell is drawn using a `<rect>` element. The x- and y-positions of the cells are mapped with `scaleBand` axes to align factors and disease outcomes. Each cell's color represents the magnitude and direction of the correlation, calculated earlier, and is assigned using a sequential color scale: `d3.scaleSequential(d3.interpolateRdBu)`. Numeric labels are added with `svg.selectAll("text")`, formatted using `d3.format(".2f")` to ensure precision and readability. A horizontal color legend is constructed manually within the SVG to guide interpretation of the scale, enhancing visual clarity of the data relationships.

Chart 2 (RQ2): Bar Charts of Lifestyle Factors and Disease

These correlation matrices are rendered as heatmaps using `d3.create("svg")`, where each cell is drawn using a `<rect>` element. The x- and y-positions of the cells are mapped with `scaleBand` axes to align factors and disease outcomes. Each cell's color represents the magnitude and direction of the correlation, calculated earlier, and is assigned using a sequential color scale: `d3.scaleSequential(d3.interpolateRdBu)`. Numeric labels are added with `svg.selectAll("text")`, formatted using `d3.format(".2f")` to ensure precision and readability. A horizontal color legend is constructed manually within the SVG to guide interpretation of the scale, enhancing visual clarity of the data relationships.

Chart 3 (RQ3): Bar Chart of Demographic Filter and Prevalence

Disease prevalence is visualized using Observable Plot's `Plot.barY()` function. The user can apply demographic filters using custom checkboxes (implemented via `exclusiveCheckboxInput()`), and filter logic is applied in `filteredData`. When the user selects a grouping variable such as "Race" via `stackBy`, the bar chart is updated to display grouped or stacked bars, with colors and legends adapting accordingly.

Chart 4 Pt. 2 (RQ4 Pt. 2): Disease Co-Occurrence Tree

The tree diagram is constructed using `d3.hierarchy()` and laid out horizontally with `d3.tree().nodeSize([dx, dy])`. SVG path elements draw curved lines between parent and child nodes using `d3.linkHorizontal()`, and disease nodes are represented as circles with text labels. To improve legibility, each text label is cloned with a white stroke layer to ensure contrast against the background. The `viewBox` of the SVG is computed dynamically to ensure the layout remains responsive regardless of the depth of the tree.

Chart 5 (RQ5): Sleep Time Comparison

The chart of sleep time differences is created in `chartGrid()` using `d3.create("svg")` and manual layout logic to render three subplots side by side. Each subplot shows two bars for "Yes" and "No" groups, using separate `scaleBand()` and `scaleLinear()` axes. Tooltips are implemented using an appended `DOM.element("div")` tooltip and `mouseover/mousemove` event handlers. A floating label (`hoverLabelGroup`) displays values above each bar, while statistical summaries are displayed below the plot using the `summary` cell. This summary

evaluates the corresponding p-value from `statsData` and dynamically generates a textual conclusion based on the selected disease in the `diseaseSelect` dropdown.

The code for the visualizations resides in Observable notebooks, which are embedded into the website using `<iframe>` elements within `viz.html`. While embedding via `<iframe>` ensures easy integration, it introduces height management challenges. To address this, we used Observable’s built-in `resize-observer` script, which is imported via <https://static.observableusercontent.com/resize-observer/3.2.0/auto.js> to auto-adjust `iframe` height dynamically based on the web browser’s window size.

These notebooks are powered by JavaScript, using libraries like D3.js and Observable Plot to construct the data charts. Interactivity is achieved through Observable’s reactive programming model, where user inputs such as checkboxes and dropdowns are defined using input views that automatically trigger updates to dependent cells. Data is typically loaded using `d3.csv()` or Observable’s built-in file attachments.

Shared cells such as `filteredData`, `getSelectedValues()`, or `stackBy` apply filters to the dataset, and the updated data is passed into chart-rendering functions. This modular setup allows visualizations to be developed and maintained independently, and then seamlessly embedded into the website without needing to re-implement chart logic in the HTML files. In some cases, statistical results are dynamically displayed based on the outputs produced from the Python-based analysis.

Overall, this design supports scalability and ease of integration. New visualizations can be easily added by building them independently in Observable and then connecting them to the existing page structure via `<iframe>`. The result is a maintainable and dynamic webpage that bridges backend analysis with front-end interactivity.

10 Evaluation

The entire website is currently live and publicly accessible, with the majority of the features functioning as intended. All primary navigation links work correctly, and the interactive data visualizations update dynamically based on user input. All of the javascript code within the observable notebooks creating the visualizations are working as well. The survey serves as the main method of evaluation, offering users the option to submit real-time and personalized feedback. User inputs are collected and are then used to improve the website even further. The emphasis here is not just the correctness of the graphs, but also their aesthetics, presentation, and clarity. Furthermore, we also collect data from the prediction tool, using them to improve the underlying machine learning algorithms for better and more accurate predictions.

11 Future Work

While this project presents a comprehensive analysis of chronic disease indicators and their relationships with demographic and lifestyle factors, several opportunities remain for future work. For example, future studies could incorporate longitudinal or time-series data to track changes in health behaviors and disease progression over time. This would support more robust causal inferences and better inform prevention strategies. Additionally, integrating more advanced machine learning models (e.g., random forests, gradient boosting, or neural

networks) could improve predictive accuracy and uncover nonlinear relationships between variables. To increase external validity, findings from the current dataset could be compared against other health datasets (e.g., NHANES, MIMIC) or validated with clinical data, ensuring the robustness and generalizability of observed correlations. By addressing these directions, this research can evolve into a more powerful tool for understanding, predicting, and ultimately reducing the burden of chronic disease on both individual and population levels.

12 Group Members Work Allocations

- **Ayaan Omair**

- Developed the website skeleton and blueprint and created the navbar
- Worked on creating the home page of the website
- Created website aesthetic
- Created visualization (Chart 2)
- Report writing (Reflection section, Requirement Analysis)
- Survey creation and survey maintenance
- Data harvesting from survey

- **Lauren Fuller**

- Made the formatting and aesthetics of the website consistent across all pages through CSS files
- Embedded the visualization charts into the website in a compatible format
- Created two visualizations (Chart 3 - interactive bar chart & Chart 4 Pt. 2 - tree diagram)
- Contributed to writing approximately three sections of the report
- Deployed the website through the GitHub Pages server
- Maintained GitHub repository storing all relevant files

- **Phuc Nguyen**

- Calculated correlation data needed for Charts 1 and 4, performing data analysis in Python
- Created visualizations to answer research questions 1 and 4
- Report writing (Related Work and Evaluation sections)
- Creation of the disease predictor tool

- **Hayley Hawkins**

- Conducted statistical analysis for Chart 5
- Created visualization for Chart 5
- Wrote approximately four sections of the report (Introduction, Contributions, Requirement Analysis, Future Work)
- Created the final presentation and contributed all but two slides

13 Reflection

As Data Science majors, we have a strong foundation in building machine learning models, analyzing complex datasets, and assessing model performance. While these are central to our academic experience, this project challenged us to go beyond traditional analysis and explore new ways to present our findings. Rather than relying solely on static charts or written summaries, we learned how to use JavaScript to create dynamic, interactive visualizations that make our results more intuitive and accessible to a broader audience. In addition, we gained hands-on experience in front-end web development by building a website using HTML and CSS. This website serves as a platform to showcase our research questions, visualizations, and predictive tools in a cohesive and visually appealing way. Through this process, we discovered the value of combining data science with web development to create more engaging and accessible presentations of our work. The skills we acquired—specifically in HTML, CSS, and JavaScript—not only enhanced this project but also open up future opportunities. For example, we now have the foundation to build personal portfolios that highlight our data science projects, showcase our code and results, and demonstrate our web development capabilities to potential employers or collaborators. This project ultimately broadened our technical toolkit and showed us how impactful data science can be when paired with thoughtful design and communication.

14 References:

1. Pytlak, K. *Personal Key Indicators of Heart Disease* [Internet]. Kaggle; 2022 [cited 2025 Apr 15]. Available from: <https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease>
2. Turk, A., Kunej, T. Shared Genetic Risk Factors Between Cancer and Cardiovascular Diseases. *Frontiers in Cardiovascular Medicine*. 2022 Jul 7; 9:931917. doi: 10.3389/fcvm.2022.931917. PMID: 35872888; PMCID: PMC9300967.
3. Ghodeshwar, G.K., Dube, A., Khobragade, D. Impact of Lifestyle Modifications on Cardiovascular Health: A Narrative Review. *Cureus*. 2023 Jul 28; 15(7):e42616. doi: 10.7759/cureus.42616. PMID: 37641769; PMCID: PMC10460604.
4. Ambrish, G., Ganesh, B., Ganesh, A., Srinivas, C., Dhanraj, and Mensinkal, K. *Logistic Regression Technique for Prediction of Cardiovascular Disease*. *Global Transitions Proceedings*. 2022; 3(1): 127–130.
5. **Our Website:** <https://lauren-fuller.github.io/679-Final-Project-Webpage/home.html>